C PROGRAMMING ASSESSMENT 2

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Q1 – Precedence, Associativity and the Discount Bug

```
#include <stdio.h>
int main()
{
   double price, qty, gross, discount, tax, total, net;
   int senior;
   printf("Enter price and quantity: ");
   scanf("%lf %lf", &price, &qty);
   printf("Senior citizen? (1-Yes, 0-No): ");
   scanf("%d", &senior);
   gross = price * qty;
   discount = gross * 15 / 100 * senior;
   tax = (gross - discount) * 5 / 100;
   total = gross - discount + tax;
   net = total > 1000 ? total - 50 : total;
   printf("Gross Amount = %.2lf\n", gross);
   printf("Discount = %.2lf\n", discount);
   printf("GST = %.2lf\n", tax);
   printf("Total Bill = %.2lf\n", total);
   printf("Net Amount = %.2lf\n", net);
   return 0;
}
```

OUTPUT :

```
Enter price and quantity: 100
5
Senior citizen? (1-Yes, 0-No): 1
Gross Amount = 500.00
Discount = 75.00
GST = 21.25
Total Bill = 446.25
Net Amount = 446.25
```

Q2 – Short-Circuit Evaluation

```
#include <stdio.h>
int calls = 0;
int verify_card()
{
   calls++;
   return 1;
}
int main()
{
   int senior = 700;
   int eligible = 600;
   int disability = 300;
   calls = 0;
   for (int i = 0; i < 1000; i++)
   {
      if ((i < senior && i < eligible) || verify_card())
      {
      }
   }
   printf("Order 1 calls = %d\n", calls);
   calls = 0;
   for (int i = 0; i < 1000; i++)
   {
      if (verify_card() || (i < senior && i < eligible))
      {
      }
   }
   printf("Order 2 calls = %d\n", calls);
   return 0;
}
```

OUTPUT :

```
Order 1 calls = 400
Order 2 calls = 1000
```

Q3 – Bitwise Operators

```
#include <stdio.h>
enum { LIGHT = 0, FAN = 1, AC = 2, HEATER = 3, PUMP = 4 };
unsigned char switch_on(unsigned char reg, int dev)
{
   return reg | (1u << dev);
}
unsigned char switch_off(unsigned char reg, int dev)
{
   return reg & ~(1u << dev);
}
unsigned char toggle(unsigned char reg, int dev)
{
   return reg ^ (1u << dev);
}
int is_on(unsigned char reg, int dev)
{
   return (reg & (1u << dev)) != 0;
}
void show(unsigned char reg)
{
   for (int bit = 7; bit >= 0; bit--)
      putchar(((reg >> bit) & 1u) ? '1' : '0');

   printf("\n");
}
int main()
{
   unsigned char reg = 0;
   reg = switch_on(reg, LIGHT);
   reg = switch_on(reg, AC);
   show(reg);
```

```
    reg = toggle(reg, FAN);
    reg = switch_off(reg, LIGHT);
    show(reg);

    printf("AC is %s, PUMP is %s\n",
        is_on(reg, AC) ? "ON" : "OFF",
        is_on(reg, PUMP) ? "ON" : "OFF");

    return 0;
}
```

OUTPUT :

```
00000101
00000110
AC is ON, PUMP is OFF
```

Q4 – else-if Ladder or switch

```
#include <stdio.h>
char grade_if(int marks)
{
    if (marks < 0 || marks > 100)
        return '?';
    else if (marks >= 90)
        return 'S';
    else if (marks >= 80)
        return 'A';
    else if (marks >= 70)
        return 'B';
    else if (marks >= 60)
        return 'C';
    else if (marks >= 50)
        return 'D';
    else
        return 'F';
}

char grade_switch(int marks)
{
    if (marks < 0 || marks > 100)
        return '?';
    switch (marks / 10)
    {
        case 10:
        case 9: return 'S';
        case 8: return 'A';
        case 7: return 'B';
        case 6: return 'C';
        case 5: return 'D';
```

```
        default: return 'F';
    }
}
int main()
{
    int marks[] = {100, 95, 83, 50, 49, -5};

    for (int i = 0; i < 6; i++)
    {
        printf("%d: %c  %c\n",
            marks[i],
            grade_if(marks[i]),
            grade_switch(marks[i]));
    }

    return 0;
}
```

OUTPUT

```
100: S  S
95: S  S
83: A  A
50: D  D
49: F  F
-5: ?  ?
```

Q5 – Count the Work Done by a Loop

```
#include <stdio.h>
unsigned long checks;
int prime_for(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    for (unsigned long i = 2; i <= n / i; i++)
    {
```

```
  checks++;
        if (n % i == 0)
            return 0;
    }
    return 1;
}
int prime_while(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    unsigned long i = 2;
    while (i <= n / i)
    {
        checks++;
        if (n % i == 0)
            return 0;
        i++;
    }
    return 1;
}
int prime_do(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    unsigned long i = 2;
    do
    {
        if (i > n / i)
            break;
        checks++;
        if (n % i == 0)
            return 0;
```

```
        i++;
    } while (1);
    return 1;
}
int main()
{
    unsigned long numbers[] = {97, 100, 7919, 1000003};
    for (int i = 0; i < 4; i++)
    {
        int result = prime_for(numbers[i]);
        printf("%lu: %s, divisions = %lu\n",
            numbers[i],
            result ? "Prime" : "Not Prime",
            checks);
    }
    return 0;
}
```

OUTPUT :

```
97: Prime, divisions = 8
100: Not Prime, divisions = 1
7919: Prime, divisions = 87
1000003: Prime, divisions = 999
```

Q6 – Loan Repayment Schedule

```
#include <stdio.h>
int main()
{
    double balance, rate, payment;
    double interest, total_interest = 0;
    int month = 0;
    printf("Enter loan amount: ");
    scanf("%lf", &balance);
```

```
printf("Enter monthly interest rate (%%): ");
scanf("%lf", &rate);
printf("Enter monthly repayment: ");
scanf("%lf", &payment);

if (balance <= 0 || payment <= 0 || rate < 0)
{
   printf("Invalid input\n");
   return 0;
}

while (balance > 0.005 && month < 360)
{
   interest = balance * rate / 100;
   total_interest += interest;
   balance += interest;
   if (payment <= interest)
   {
      printf("Loan cannot be repaid with this amount.\n");
      return 0;
   }

   if (balance < payment)
      balance = 0;
   else
      balance -= payment;
   month++;
   printf("Month %d: Balance = %.2lf\n",
      month, balance);
}
```

```
    if (balance <= 0.005)
    {
        printf("Loan completed in %d months\n", month);
        printf("Total Interest = Rs. %.2lf\n", total_interest);
    }
    else
    {
        printf("Maximum repayment period reached.\n");
    }

    return 0;
}
```

OUTPUT :

```
Enter loan amount: 10000
Enter monthly interest rate (%): 2
Enter monthly repayment: 2000
Month 1: Balance = 8200.00
Month 2: Balance = 6364.00
Month 3: Balance = 4491.28
Month 4: Balance = 2581.11
Month 5: Balance = 632.73
Month 6: Balance = 0.00
Loan completed in 6 months
Total Interest = Rs. 645.38
```

Q7 – Diagnose, Redesign and Validate

```
#include <stdio.h>
int main()
{
    unsigned long n, i;
    int prime = 1;
    int checks = 0;
    printf("Enter a number: ");
    scanf("%lu", &n);
```

```
    if (n < 2)
        prime = 0;
    else if (n == 2)
        prime = 1;
    else if (n % 2 == 0)
        prime = 0;
    else
    {
        for (i = 3; i <= n / i; i += 2)
        {
            checks++;


            if (n % i == 0)
            {
                prime = 0;
                break;
            }
        }
    }
    if (prime)
        printf("%lu is Prime\n", n);
    else
        printf("%lu is Not Prime\n", n);
    printf("Number of divisions = %d\n", checks);
    return 0;
}
```

OUTPUT :

```
Enter a number: 100003
100003 is Prime
Number of divisions = 157
```